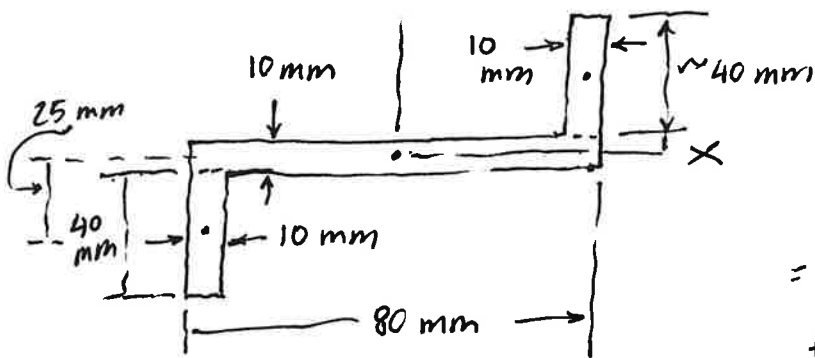


Ex Problem



Find: $I_x, I_y, I_{xy}, \theta_p, I_{MIN}, I_{MAX}$

$$I_x = \sum (\bar{I} + Ad^2)$$

$$= \left(\frac{1}{12} (80)(10)^3 + 0 \right) + 2 \left(\frac{1}{12} (10)(40)^3 + (10)(40)(25)^2 \right)$$

$$I_x = 6.67 \cdot 10^3 + 2(303.3 \cdot 10^3)$$

$$I_x = 0.6133 \cdot 10^6 \text{ mm}^4$$

$$I_y = \sum (\bar{I} + Ad^2)$$

$$= \left[\frac{1}{12} (10)(80)^3 + 0 \right] + 2 \left[\frac{1}{12} (40)(10)^3 + (40)(10)(35)^2 \right]$$

$$I_y = 426.7 \cdot 10^3 + 2(493.3 \cdot 10^3)$$

$$I_y = 1.413 \cdot 10^6 \text{ mm}^4$$

$$I_{xy} = \sum (\bar{I} + Ad_x d_y)$$

$$= [0 + 0]$$

$$+ [0 + (10)(40)(+35)(+25)]$$

$$+ [0 + (10)(40)(-35)(-25)]$$

$$I_{xy} = 0.700 \cdot 10^6 \text{ mm}^4$$

$$I_{AVG} = \frac{0.6133 + 1.413}{2} = 1.013 \cdot 10^6 \text{ mm}^4$$

$$\text{Radius} = \sqrt{\left(\frac{I_x - I_y}{2} \right)^2 + I_{xy}^2} = 0.806 \cdot 10^6$$

$$I_{MAX} = 1.01 + .806$$

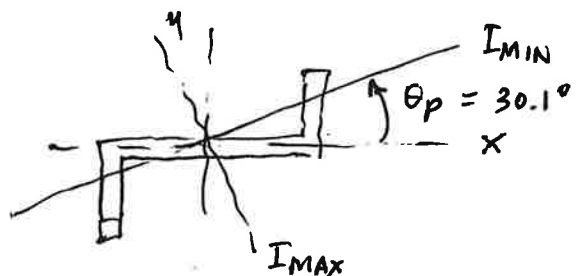
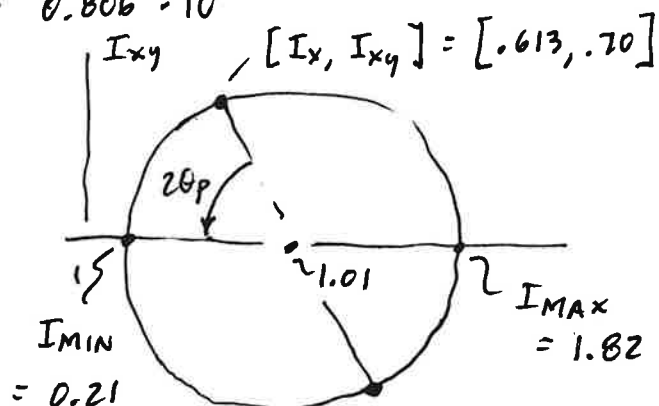
$$= 1.82 \cdot 10^6 \text{ mm}^4$$

$$I_{MIN} = 1.01 - .806$$

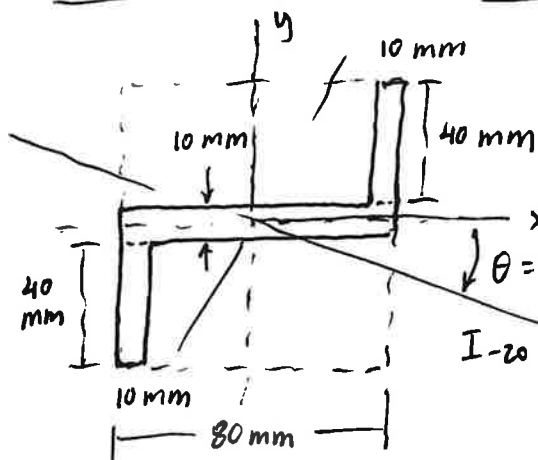
$$= 0.21 \cdot 10^6 \text{ mm}^4$$

$$2\theta_p = \tan^{-1} \frac{-I_{xy}}{\left(\frac{I_x - I_y}{2} \right)} = 60.3^\circ$$

$$\theta_p = 30.1^\circ$$



Ex Problem : I_{70} I_{θ} , $I_{\theta+90}$, I_{xy} at Given θ



$$I_x = .613 \cdot 10^6 \text{ mm}^4$$

$$I_y = 1.413 \cdot 10^6 \text{ mm}^4$$

$$I_{xy} = .70 \cdot 10^6 \text{ mm}^4$$

Already found: $\theta_p = 30.1^\circ$

$$I_{\max} = 1.82, I_{\min} = 0.21$$

At $\theta = -20^\circ$, please determine: I_u , I_v , I_{uv} (I_{θ} , $I_{\theta+90}$, I_{uv})

By these methods:

① 2 θ equations

② θ equations

③ Mohr's Circle

① 2 θ Equations

$$I_{-20} = I_u = \frac{.613 + 1.413}{2} + \frac{(.613 - 1.413)}{2} \cos 2(-20) - (.70) \sin 2(-20)$$

Use same equation but

$\theta = 70$

$= -20 + 90$

$$I_{-20} = I_u = 1.16 \cdot 10^6 \text{ mm}^4$$

Checks w/ Excel!

$$I_v = I_{70} = () + () \cos 2(70) - (.7) \sin 2(70)$$

$$I_v = I_{70} = .869 \cdot 10^6 \text{ mm}^4$$

Checks w/ Excel!

$$I_{uv} = \frac{.613 - 1.413}{2} \sin 2(-20) + (.7) \cos 2(-20)$$

$$I_{uv} = 0.793 \cdot 10^6 \text{ mm}^4$$

② θ Equations

$$I_{-20} = I_u = .613 \cos^2(-20) + 1.413 \sin^2(-20) - 2(.7) \sin(-20) \cos(-20)$$

$$\boxed{I_{-20} = I_u = 1.16 \cdot 10^6 \text{ mm}^4}$$

$$I_{70} = I_v = .613 \cos^2(70) + 1.413 \sin^2(70) - 2(.7) \sin 70 \cos 70$$

$$\boxed{I_{70} = I_v = .869 \cdot 10^6 \text{ mm}^4}$$

I_{uv} (use the 2θ Equation -- easier)

③ Mohr's Circle

Procedure: (a) Plot MC; locate I_{avg} , I_{max} , I_{min}

(b) Plot $[I_x, I_{xy}]$ for current orientation.

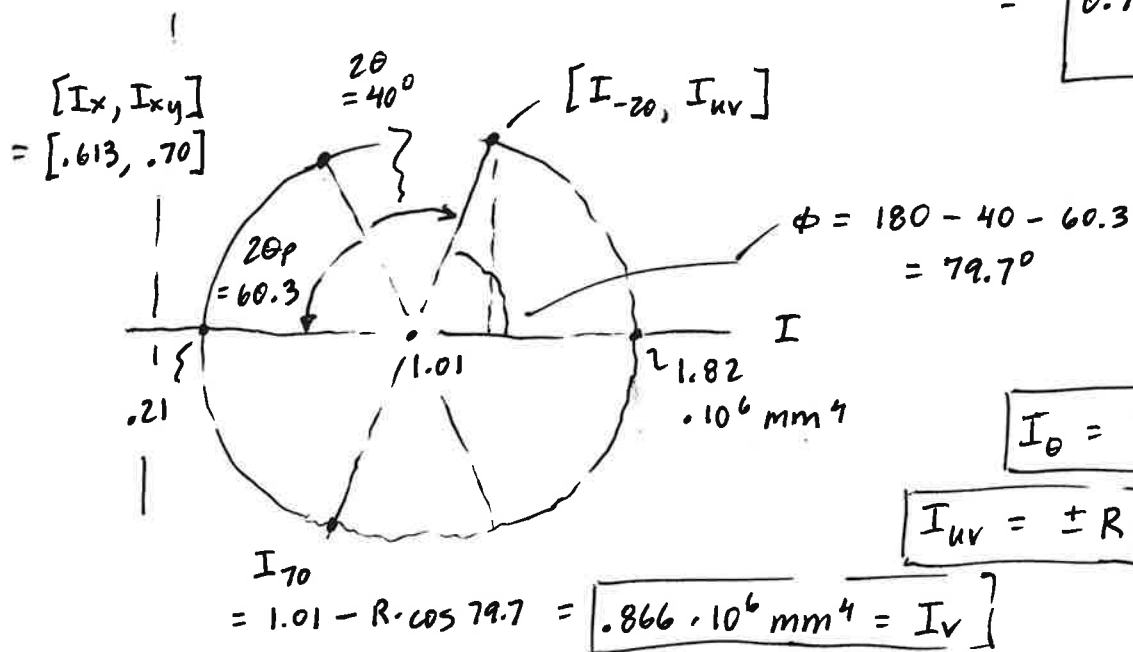
(c) Rotate line $2\theta = 2(-20) = -40$ from $[I_x, I_{xy}]$ line.

(d) Note: Radius = $0.806 \cdot 10^6 \text{ mm}^4$

(e) Determine $\phi = 79.7^\circ$ as shown.

$$(f) \quad I_{-20} = \underset{\text{"}I_{avg}\text{"}}{1.01} + \overset{R}{.806} \cos 79.7 = \boxed{1.154 \cdot 10^6 \text{ mm}^4 = I_{-20}}$$

$$(g) \quad I_{uv} = R \sin \phi = .806 \sin 79.7 = \boxed{0.793 \cdot 10^6 \text{ mm}^4 = I_{uv}}$$



$$\boxed{I_\theta = I_{avg} \pm R \cdot \cos \phi}$$

$$\boxed{I_{uv} = \pm R \cdot \sin \phi}$$

$$I_{70} = 1.01 - R \cdot \cos 79.7 = \boxed{.866 \cdot 10^6 \text{ mm}^4 = I_v}$$